IBM DATA ANALYST INTERNSHIP

PROJECT REPORT

Customer Churn Analysis and Prediction

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Domain: Data Analytics / Predictive Analytics

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

This report documents a complete data-analytics project covering data cleaning, exploratory analysis, predictive analytics, machine-learning basics, visualization, customer segmentation, and business insights.

1. Introduction

Customer churn refers to customers discontinuing a service. For subscription-based businesses, understanding churn is important because customer retention is closely connected to revenue and long-term growth. This project analyzes customer-level information and identifies factors associated with churn. The analysis also demonstrates how machine-learning techniques can be used to estimate whether a customer is likely to churn.

2. Problem Statement

The objective is to study customer behavior, identify patterns related to churn, build predictive models for churn classification, and divide customers into meaningful segments. The final analysis is intended to convert raw customer data into understandable business insights.

3. Objectives

The major objectives of the project are:

• Clean and preprocess customer data.

• Identify missing values and improve data quality.

• Perform exploratory data analysis using statistical summaries and visualizations.

• Study the relationship between churn and customer attributes.

• Build predictive models using Logistic Regression and Decision Tree.

• Evaluate models using Accuracy, Precision, Recall, F1-score and ROC-AUC.

• Perform customer segmentation using K-Means clustering.

• Generate business-oriented insights from the analysis.

4. Internship Learning Alignment

The project applies the major topics listed in the internship structure: data analytics fundamentals, data cleaning and wrangling, exploratory data analysis, predictive analytics, supervised learning, machine-learning basics, clustering, visualization, and AI-driven insights. The project therefore serves as an integrated demonstration of the skills covered during the internship.

5. Dataset

The Python program is designed to use a local file named customer_churn.csv when it is available. For independent execution without an external dataset, the program contains a reproducible demonstration dataset generator. The generated data contains customer demographics, tenure,

internet service, contract type, payment method, support services, monthly charges, total charges, and churn status. A final submission should clearly identify the actual dataset used.

6. Data Cleaning and Preprocessing

The data-cleaning stage removes duplicate records, standardizes column names, converts numerical fields to numeric data types, and handles missing values. For machine learning, numerical variables are imputed using the median and standardized. Categorical variables are imputed using the most frequent value and transformed using one-hot encoding. CustomerID is excluded from model training because it is an identifier rather than a predictive feature.

7. Exploratory Data Analysis

EDA is performed to understand the structure and distribution of the data. The project examines overall churn distribution, churn by contract type, tenure patterns, monthly charges, and correlations among numerical variables. Charts are saved automatically in the outputs folder so they can also be used in the final presentation or Power BI dashboard.

8. Predictive Analytics

Two supervised-learning models are implemented: Logistic Regression and Decision Tree Classifier. The dataset is divided into training and testing sets using stratified sampling. The preprocessing pipeline is fitted only as part of the training workflow, reducing the risk of inconsistent transformation between training and testing data.

9. Model Evaluation

The models are evaluated using Accuracy, Precision, Recall, F1-score and ROC-AUC. Accuracy describes overall classification correctness. Precision indicates how many predicted churn cases were actually churn cases. Recall indicates how many actual churn cases were detected. F1-score balances precision and recall. ROC-AUC measures the model's ability to distinguish between the two classes across classification thresholds.

10. Customer Segmentation

K-Means clustering is used as an unsupervised-learning component. Tenure, monthly charges and total charges are standardized before clustering. Several cluster counts are tested using silhouette score, and the cluster count with the highest score is selected for the demonstration analysis. This can

help a business identify groups of customers with different usage and spending characteristics.

11. Business Insights

The project focuses on practical questions such as which contract groups show greater churn, whether tenure is related to retention, how monthly charges differ between churned and retained customers, and whether support services are associated with churn. These observations can be used to design targeted retention strategies, subject to validation on real organizational data.

12. Power BI / Visualization Extension

The cleaned dataset and generated outputs can be used to build a Power BI dashboard. Recommended dashboard elements include total customers, churn rate, churn by contract, churn by payment method, monthly-charge distribution, tenure analysis, and customer segments. Interactive filters can be added for contract, internet service, payment method and customer characteristics.

13. Tools and Technologies

Python was used as the main programming language. Pandas and NumPy were used for data manipulation and numerical operations. Matplotlib and Seaborn were used for visualization. Scikit-learn was used for preprocessing, predictive modeling, evaluation and clustering. The results can additionally be presented in Power BI or Tableau.

14. Limitations

The demonstration version generates synthetic data when the original CSV is unavailable. Synthetic data is useful for demonstrating the workflow but should not be interpreted as evidence about a real company or real customers. Model performance may change substantially when evaluated on an actual organizational dataset. Real-world deployment would require validation, monitoring, privacy controls, and domain review.

15. Future Scope

Future improvements could include using a verified real-world dataset, hyperparameter tuning, cross-validation, feature importance and explainability, additional customer segments, Power BI interactivity, automated reporting, and deployment of the predictive model as an internal analytics service.

16. Conclusion

This project demonstrates an end-to-end data analytics workflow, beginning with raw customer data and progressing through cleaning, EDA, preprocessing, predictive modeling, clustering, visualization, and business interpretation. It provides a practical example of how data analytics and machine learning can support customer-retention analysis and data-driven decision making.

17. Project Deliverables

File	Purpose
customer_churn_analysis.py	Complete analysis and machine-learning code
requirements.txt	Python libraries required to run the project
README.md	Setup, execution and project documentation
Project_Report_Customer_Churn_Analysis.pdf	Formal internship project report